

SITEMind

The verification layer for data-centre EPC delivery. It reads a submittal, checks every parameter against real Indian code, and proves each call, in seconds instead of days.

Presented by **Awnikant Ajay** · Team HashForge · IIIT Kalyani

Live prototype → sitemind.awni.in

The screenshot displays the SiteMind web application interface. The left sidebar contains a 'COMMAND CENTER' menu with options: Overview (selected), Project Timeline, Compliance, Copilot, Schedule, Supply Chain, Commissioning QA, Knowledge Graph, Knowledge Base, and Codebook. The main content area is titled 'Project Overview' and includes a summary of the project: 'One AI layer over a 48 MW Chennai data-centre build — Day 175 of ~337. It reads the documents, checks them against real Indian standards, and connects what it finds across schedule, procurement and commissioning — see the whole build on the Project Timeline.' Below this, a section titled 'WHAT SITEMIND PROCESSED' shows four metrics: DOCUMENTS READ (12), CLAUSES CHECKED (12), CROSS-REFERENCES FOUND (5), and CONFLICTS SURFACED (2). The 'NEXT DECISIONS' section lists three items: 'Snagging / punch-list close-out — critical-path slip' (due now), 'DRUPS 2.5MW — confirm alternative supplier' (due Day 185), and 'LV switchgear 4000A — confirm alternative supplier' (due Day 185). The 'LATEST ON SITE' section shows a timeline of events, including 'White Space Zone B — Out Of Recommended But Within Allowable' (Day 174), 'White Space Zone C — Fail' (Day 174), 'White Space Zone D — Fail' (Day 174), 'Roof structure & cyclonic-rated cladding — predicted 5d slip' (Day 180), and 'RFI-MEP-095 — Hot-aisle containment clearance to sprinklers' (Day 145). The bottom left corner shows 'MODE' with 'Backend Live' selected.

SiteMind • Hyperscale DC — Chennai • 48 MW • Tier III

Agents online Day 175

Project Overview

SiteMind is already catching code violations, schedule slips and rework before they reach the field.

One AI layer over a 48 MW Chennai data-centre build — Day 175 of ~337. It reads the documents, checks them against real Indian standards, and connects what it finds across schedule, procurement and commissioning — [see the whole build on the Project Timeline](#).

WHAT SITEMIND PROCESSED

DOCUMENTS READ	CLAUSES CHECKED	CROSS-REFERENCES FOUND	CONFLICTS SURFACED
12	12	5	2

NEXT DECISIONS

Now	Snagging / punch-list close-out — critical-path slip +5d to project finish if unaddressed	⌵
Day 185 (10d)	DRUPS 2.5MW — confirm alternative supplier Miss this window and Uptime Bridge Rentals (interim containerized DRUPS) (+45% premium) also stops being viable	⌵
Day 185 (10d)	LV switchgear 4000A — confirm alternative supplier Miss this window and Vellore Panel Distributors (ex-stock unit) (+22% premium) also stops being viable	⌵

LATEST ON SITE

		Full timeline ⌵
Day 174	White Space Zone B — Out Of Recommended But Within Allowable	Commissioning QA
Day 174	White Space Zone C — Fail	Commissioning QA
Day 174	White Space Zone D — Fail	Commissioning QA
Day 180	Roof structure & cyclonic-rated cladding — predicted 5d slip	Schedule
Day 145	RFI-MEP-095 — Hot-aisle containment clearance to sprinklers	Copilot / RFIs

MODE
Backend Live

One missed number on a drawing can stall a billion-dollar data-centre build.

Every submittal for a hyperscale build must match hundreds of pages of structural and electrical code. Today a senior engineer checks that by hand, and a single overlooked clause means months of rework, or a facility that isn't safe to power on.

A footing poured at **30 mm** cover, against the **50 mm** legal minimum (IS 456:2000, Cl. 26.4.2.2). By the time a human catches it, the concrete has set.

Industry context, not a SiteMind metric: **67%** of data-centre EPC projects in Asia-Pacific overrun schedule by more than 10% (Turner & Townsend, 2024), led by procurement misalignment and commissioning failures.

Manual review is too slow. Generic AI tools guess.



Manual review

A senior engineer reads hundreds of pages by hand, days per document, and still misses what humans miss.



Naïve RAG chatbots

Retrieve a paragraph and let a language model *decide* if it's compliant, with no abstention gate and no guardrail against a fabricated citation.

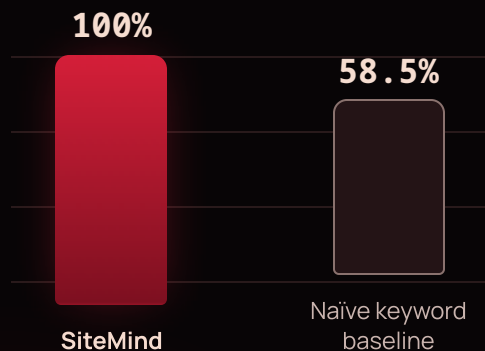


Spreadsheets & checklists

Static, don't scale to hundreds of IS clauses, and can't link an issue to the RFI, schedule task, and shipment it touches.

THE MODEL **never** DECIDES.

An LLM reads the document and writes the report. The pass-or-fail call is made by deterministic Python, against a real, cited Indian-standard clause. Auditable, reproducible, impossible to fake.



0.0%

Hallucinated citations — 32 of 32 cited clauses resolve to real IS text



21/21

Re-runnable eval scripts passing, reported per pillar

Rule-decision accuracy, macro-F1 **1.00** vs **0.56**, n=41 boundary cases
(backend/eval/report.json).

● 04 · PERCEIVE → DECIDE → EXPLAIN

Three stages. A model touches only the outer two.

1 · PERCEIVE

Read & extract

Claude Sonnet 4.6 pulls each parameter with its exact source span, behind a deterministic span-verification gate, or abstains.



2 · DECIDE

Check the clause

Deterministic Python tests the value against the cited IS/CEA clause. Re-runnable, and the one step a model may never touch.



3 · EXPLAIN

Write the finding

The model drafts the non-conformance in plain English, around a decision it did not make.

Every citation resolves to real primary-source text, digitised IS and CEA codes, never paraphrased from model memory, and each clause discloses its own reliability tier.

LLM-first where intelligence matters. Deterministic where trust matters.

THE INTELLIGENCE LAYER

Claude Sonnet 4.6, via the Claude Agent SDK. Reads and extracts from documents, answers cross-document questions, and writes every finding. This is where the product feels smart.

THE GUARANTEE LAYER

Deterministic Python. Owns the compliance verdict and resolves every citation to real clause text. A verdict or a citation can never be fabricated.

NEXT.JS 14 • TS • TAILWIND

Command Center

Overview Compliance Copilot Schedule Supply Chain Commissioning Timeline

↓ REST

FASTAPI • CLAUDE AGENT SDK

Application & Intelligence Layer

compliance copilot schedule supply_chain commissioning impact / cost

↓ MCP CLIENT

↓ HYBRID RETRIEVE

↓ GRAPH LOOKUP

FASTMCP • :8010

Codebook

search_standards get_clause 6,206 chunks

BM25 + DENSE • RRF

Knowledge Base

multi-corpus live upload abstain floor

NETWORKX

Knowledge Graph

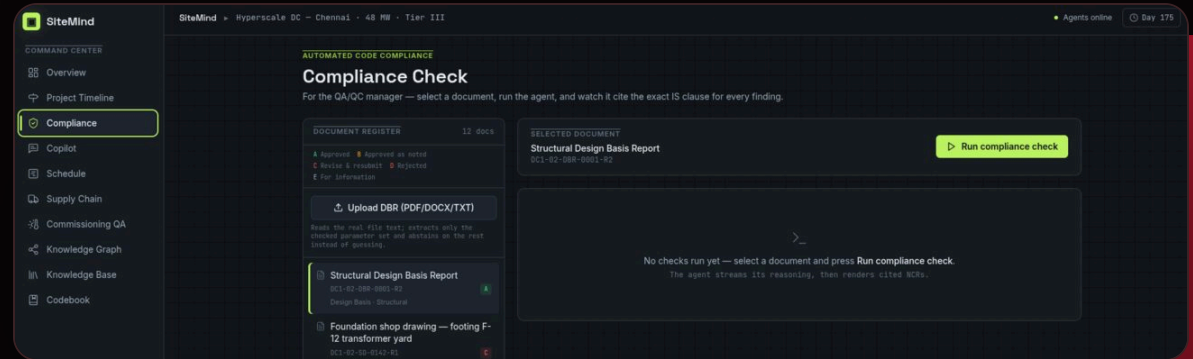
equipment ↔ spec ↔ standard ↔ RFI

No model training anywhere. The guarantee layer is offline-capable, so SiteMind stands up in secure, air-gapped data-centre environments.

Compliance, Copilot, Schedule

Compliance Agent

Catches design non-conformances before casting or procurement, cited against real IS/CEA clauses.



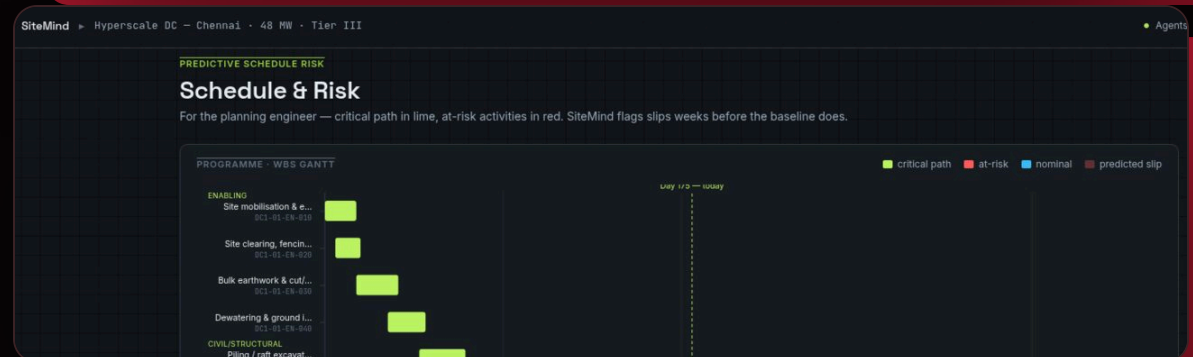
Project Copilot

Guardrailed hybrid retrieval — BM25 + dense, reciprocal-rank fused — detects previously-resolved RFIs and abstains below a deterministic floor instead of guessing.



Schedule Risk

Recomputes critical-path slip against weather and workforce, with three mitigation agents.



Supply Chain, Commissioning, Timeline

Supply Chain

Traces component latency three tiers deep, with computed viable alternatives before it hits critical path.

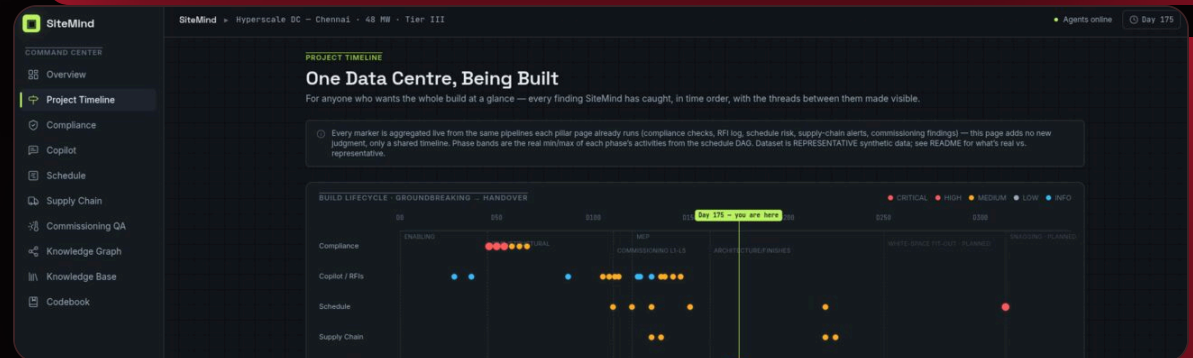
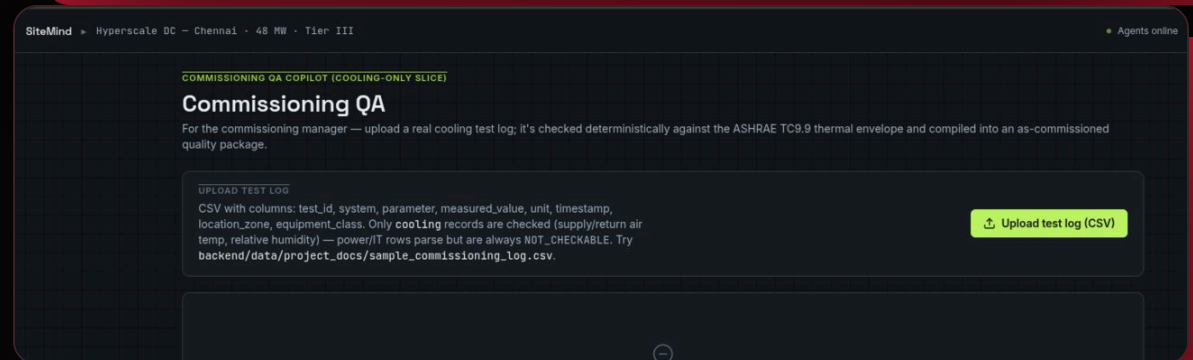
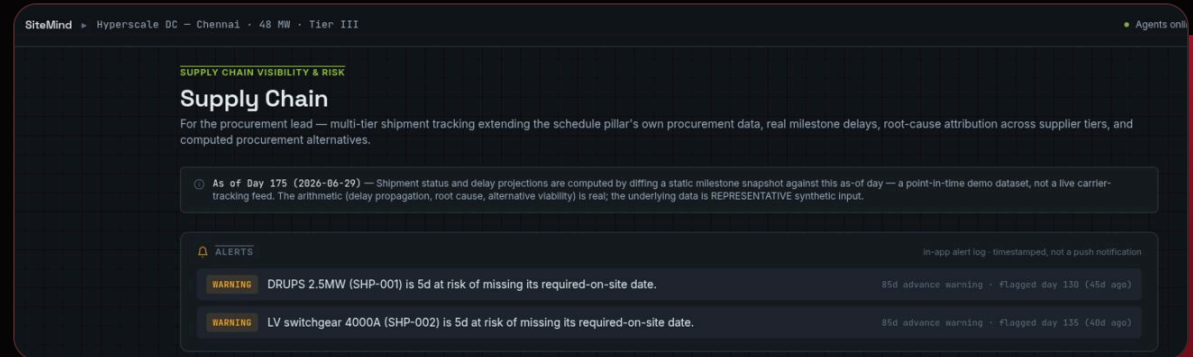
Commissioning QA

Deterministic pass / allowable / fail against the ASHRAE cooling envelope, from a real test log.

Project Timeline

Every finding, RFI, risk, and alert on one cross-linked view. Pure aggregation, zero new judgement.

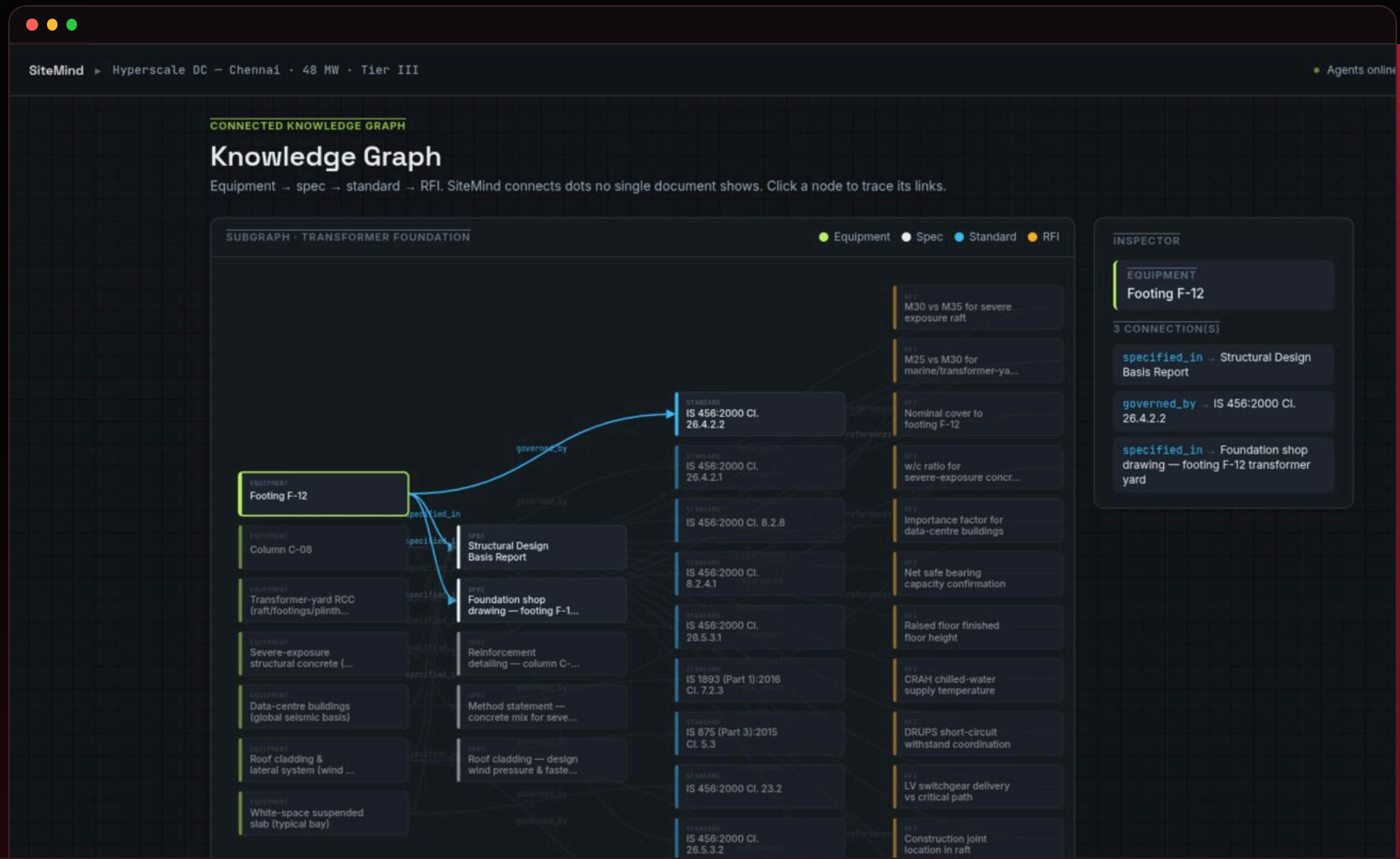
Electrical/fire commissioning is deferred on a disclosed corpus gap, not shipped as complete.



TRACED, NOT guessed.

Retrieval, where it runs, is guardrailed hybrid search with a deterministic abstention gate. But the cross-pillar memory is a NetworkX graph, not a vector store: a permanent, clickable link that traces a parameter to the physical asset, the RFI, and the schedule task it affects — never an unstable similarity guess.

Structural Submittal SHP-002 ↔ RFI-EL-112 ↔ Schedule Task DC1-04-EL-030



A verified-standards layer any agent can call.

CODEBOOK · MCP SERVICE

A standalone standards service running its own MCP server over streamable-HTTP, exposing `search_standards`, `get_clause`, and `list_corpora`. SiteMind consumes it as a client — and so can any other tool in an EPC stack.

6,206 verbatim-verified chunks · 17 real Indian codes · a source-type reliability tier on every clause.

KNOWLEDGE BASE · HYBRID RETRIEVAL

Guardrailed retrieval, not naïve RAG. BM25 keyword fused with dense embeddings by reciprocal-rank fusion, with a deterministic cosine floor as the abstention gate — the same discipline the Copilot runs on.

Multi-corpus, with live document upload, so a company's own submittals become searchable under the same cited, abstaining rules.

The middle stays deterministic: retrieval fuses and ranks candidates, but the go/no-go to answer is a fixed threshold — so the model never decides whether it has enough to speak.

Value measured in hours. Breadth added by clauses.

TRANSPARENT ECONOMICS

~174 hrs

Engineer-hours recovered on this 48 MW build, computed live

₹1.06 cr

Rework exposure avoided on the same build

21/21

Real eval scripts passing, full suite

Hours, not percentages: the Overview sums issues caught × disclosed per-issue constants (~20 hrs, ₹15L each) — labelled assumptions, not measured outcomes from a live project.

SCALES WITHOUT RETRAINING

Every check maps to a real digitised clause, so breadth grows by **adding a clause, never by retraining a model**. No training run, no dataset to relabel, no drift.

Backbone: **6,206** verbatim-verified chunks across 17 real Indian codes, exposed over MCP for any agent to query.

IS 875

IS 13920

IS 800

NBC 2016

The verification layer for every data centre built in India.

- ◆ **Widen the code library** to IS 875, IS 13920, IS 800, and NBC 2016, the most-cited code in real permitting.
- ◆ **Extend perception to drawings** with vision models, so shop drawings read the same way the text path does.
- ◆ **One backbone, many tools.** The standards service speaks MCP, so any tool in an EPC stack can call the same verified codes.

21/21 real evals passing today · legal codes and checking logic are 100% genuine ·
project data is representative, modelled on public tenders · disclosed, not hidden.

• SITEMIND • HASHFORGE

THANK *you*

It doesn't ask you to trust an AI. It shows its work, cites its source, and lets you check every call it makes. That is what makes it ready for infrastructure with zero tolerance for error.

[Launch the console → sitemind.awni.in](https://sitemind.awni.in)